owed

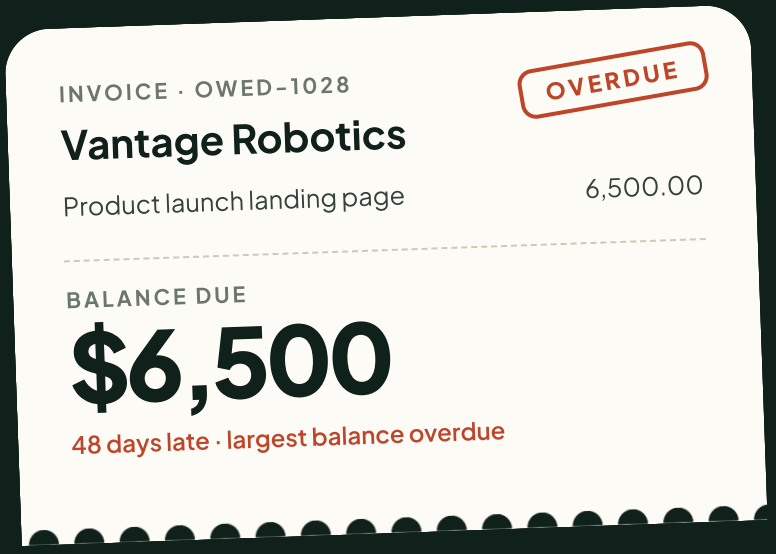
PAYPAL AI HACKATHON 2026 · PROJECT BRIEF

Get paid for the work you already did.

Owed is an AI collections copilot for freelancers and small studios who bill through PayPal. Claude agents read your invoices, learn how each client really pays, and draft the right reminder. **Nothing is sent until you approve it.**

PayPal Invoicing v2 · Transaction Search · PayPal Agent Toolkit · Claude · AG Studio Agent Framework

Live app	owed.edilsonsantos.co.za
Source (MIT)	github.com/edilsonsantos-co-za/owed





BUILT BY
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THE PROBLEM

Chasing money is draining, and one note doesn't fit every client.

For freelancers and small studios

The work is delivered, the invoice went out through PayPal, and three weeks later it is still unpaid. Do you nudge? How hard? Will a firm email cost you the client? Most people put it off, or send everyone the same polite note.

Clients are not all the same

One always pays two days early. Another pays seven weeks late, every time. The right reminder depends on the client's history, and that history is already sitting in PayPal.

HOW OWED WORKS

01 · READ

PayPal invoices

Invoicing v2 and Transaction Search become a live receivables model: outstanding, overdue, aging, and when the cash should really land.

→

02 · DRAFT

Claude agents

A collections agent judges each client from their history and drafts a friendly, firm or final reminder, with a one-line reason.

→

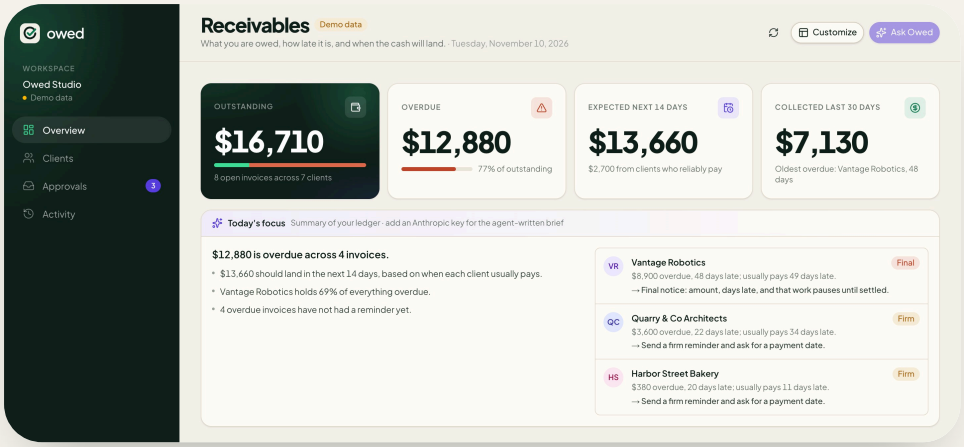
03 · APPROVE

You decide

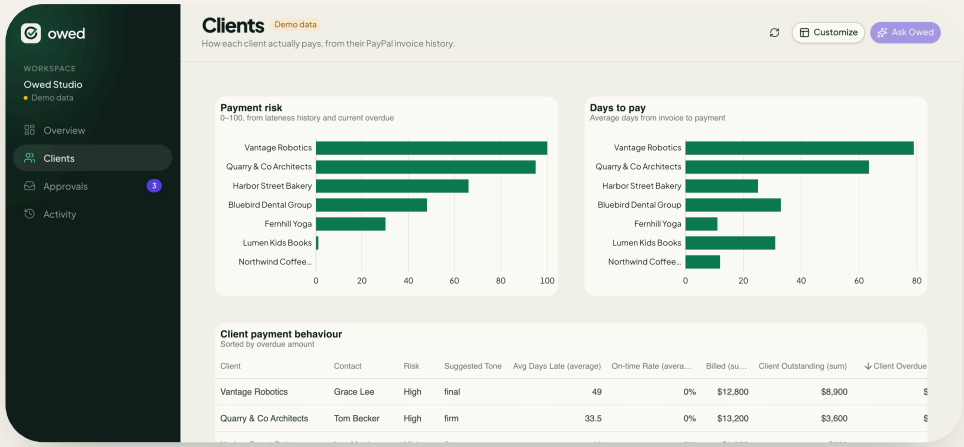
Every draft waits in Approvals. Edit, approve, and Owed sends a real PayPal reminder with a pay button. Or discard it.

THE PRODUCT

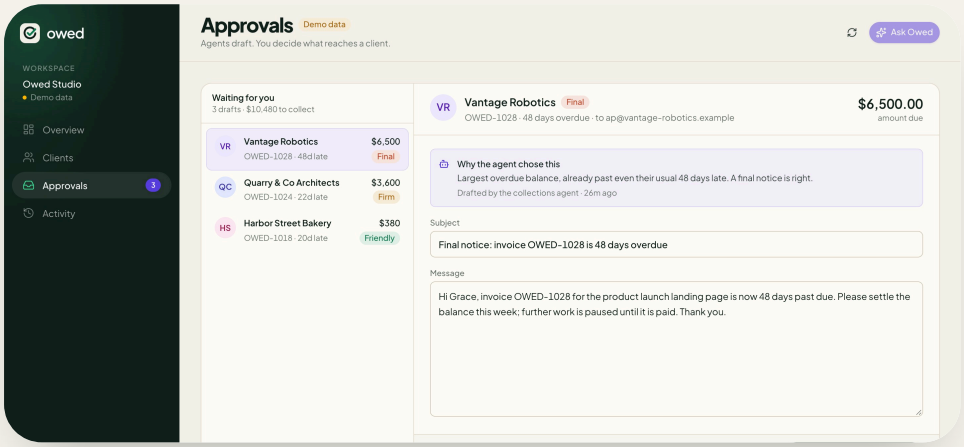
A complete app, not a proof of concept.



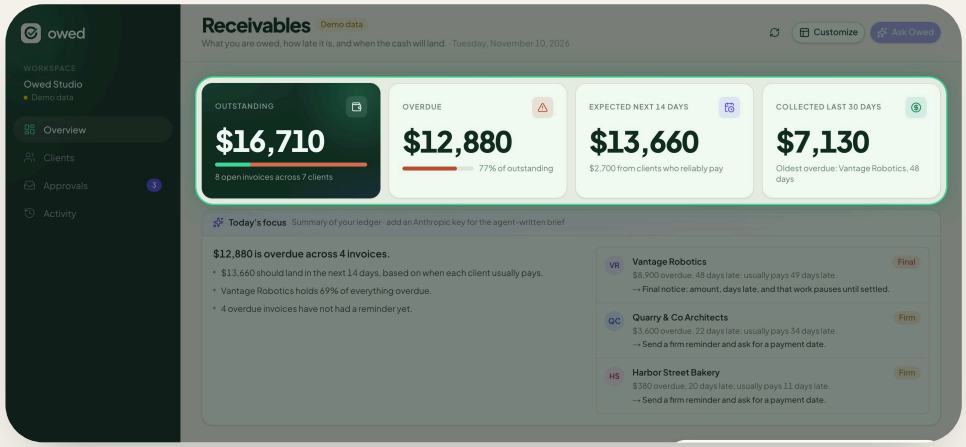
Overview. KPIs, today's focus, aging and a cash forecast shifted by each client's real habits.



Clients. Payment profile and risk score for every client, from their invoice history.



Approvals. Agent drafts with tone and reasoning; edit, approve and send, or discard.



Walkthrough. First-time users are guided page by page, with each part spotlighted.

HOW IT'S BUILT

Agents that work as a team, and can't send on their own.

Dashboard and chat	AG Studio embedded dashboards, themed to Owed, with a full-screen Customize editor. The Studio chat runs six agents in the browser.
Custom Claude adapter	Translates AG Studio's Agent Framework turns into streaming Claude calls (Anthropic Messages API, adaptive thinking), and Claude's stream back into AG-UI events.
Collections agent	Owed's own agent, which Studio's built-in lead can delegate to. Tools: `get_receivables`, `get_client_profile`, `draft_reminder`, `draft_invoice`.
Morning brief	A read-only Claude agent using the PayPal Agent Toolkit (`list_invoices`, `search_invoices`, `list_transactions`, `get_merchant_insights`).
PayPal	Invoicing v2 (list, get, create, send, remind, record payment), Transaction Search.
App	Next.js 16, React, TypeScript, Tailwind CSS v4; Logto sign-in (OIDC + PKCE); Docker, deployed on every push.

SAFETY BY CONSTRUCTION

Approval is in the code, not the prompt.

Agent tools can only **queue drafts**. The only code path that calls PayPal's `send` or `remind` endpoints is the Approve button, which needs a signed-in person.

A prompt can be talked out of a rule; a missing tool can't. Every draft, approval and PayPal call is written to the Activity log with who did it: you, an agent, or the system.

The hosted demo also caps AI turns per visitor per day.

THE PAYMENT-BEHAVIOUR MODEL

Small, explainable numbers a freelancer can check.

Average lateness for client c over settled invoices S_c

$$\bar{L}_c = \frac{1}{|S_c|} \sum_{i \in S_c} \left(t_i^{\text{paid}} - t_i^{\text{due}} \right)$$

On-time rate: paid within two days of the due date

$$r_c = \frac{\left| \{ \, i \in S_c : t_i^{\text{paid}} - t_i^{\text{due}} \le 2 \, \} \right|}{|S_c|}$$

Risk score, with W_c the days overdue of the worst open invoice

$$\text{risk}_c = \min\Big(100,\ 1.2\,\bar{L}_c + 0.9\,W_c + 25\,(1 - r_c) + 10 \cdot 1[\text{overdue}_c > 0]\Big)$$

Cash forecast: an open invoice is expected on

$$\hat{t}_i = \max\left(t_i^{\text{due}} + \bar{L}_c,\ t_{\text{today}} + 3\right)$$

Confidence falls as risk rises (low risk → high confidence).

Tone of each reminder

Final	worst invoice more than 45 days overdue
Firm	more than 20 days overdue, or a high-risk client
Friendly	everyone else

The agent explains its choice in plain words on every draft, so you can see why before you approve.

TRY IT

Live in a minute, with or without keys.

Hosted demo

Open **owed.edilsonsantos.co.za**. The landing page is public; create a free account (or use the test account in the Devpost testing notes) to open the dashboard. A first-visit walkthrough shows every page.

What to look at

- Overview: KPIs, today's focus and the cash forecast
- Clients: payment profiles and risk
- Ask Owed: "Chase everyone more than 30 days late"
- Approvals: edit a draft, approve or discard it
- Activity: who did what

Run it locally

```
git clone https://github.com/edilsonsantos-co-za/owed
cd owed
npm install
npm run dev        # http://localhost:3000
```

With no keys it runs on a built-in demo business (seven clients, each with their own payment habits). Add keys in `.env.local` to switch on each part:

`ANTHROPIC_API_KEY`	Ask Owed, the agents and the morning brief
`PAYPAL_CLIENT_ID` + `SECRET`	live sandbox data; then `npm run seed`
`AG_STUDIO_LICENSE_KEY`	removes the trial watermark

The demo video's voice-over was generated with AI tools. Screens use demo data. Open source under the MIT licence.